GROUP 7 - Project Documentation

Members:

Ballesteros, Miguel Russel

Mangubat, Harold

Teoxon, Anthon James

1. Introduction

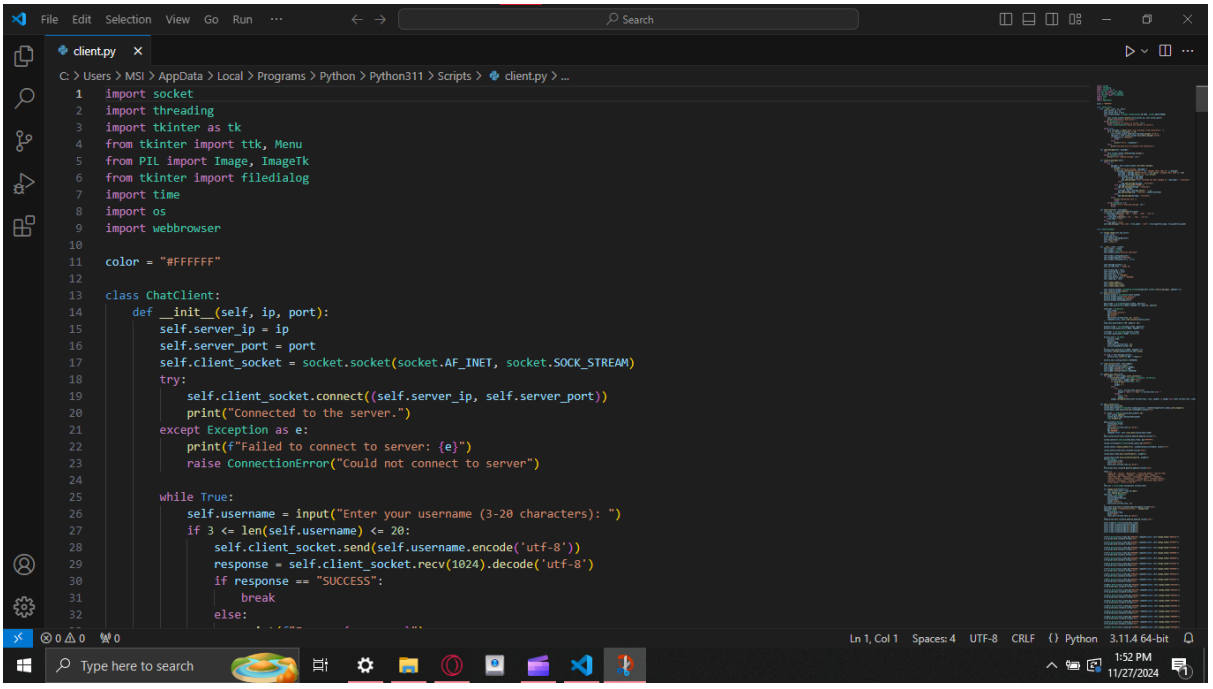
This is a human-to-human chatroom application. Our software system operates with a graphical user interface (GUI), allowing multiple users to join and communicate through a centralized server.

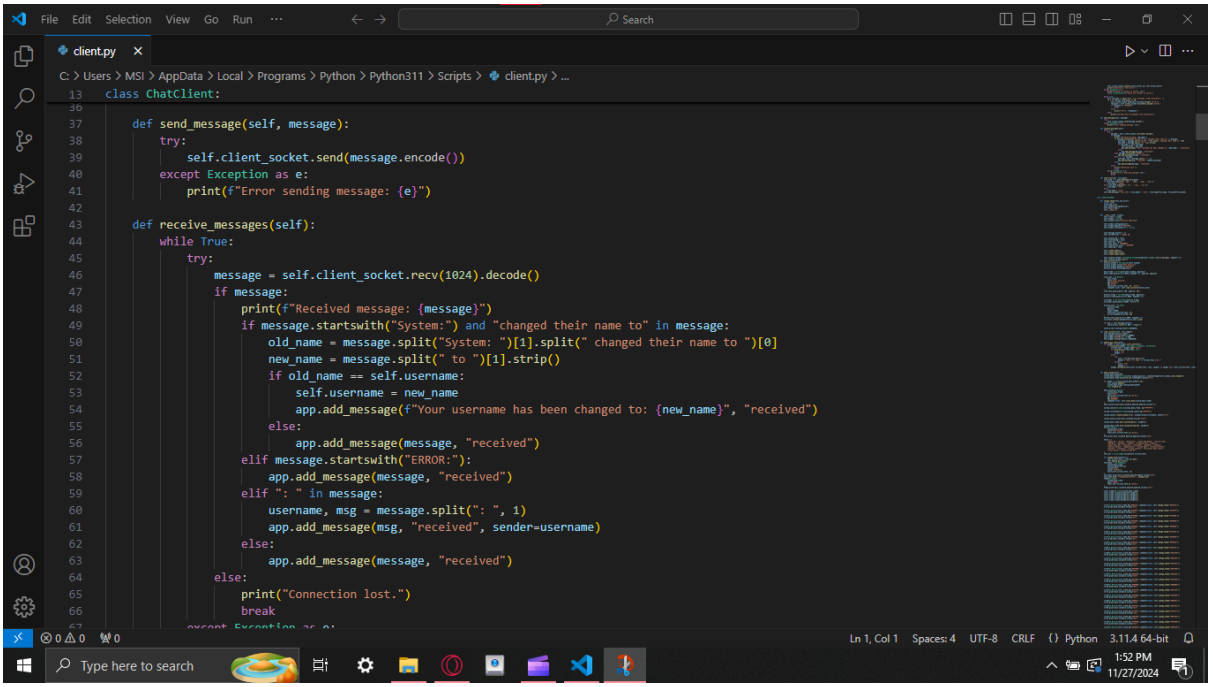
2. Objectives of the Project

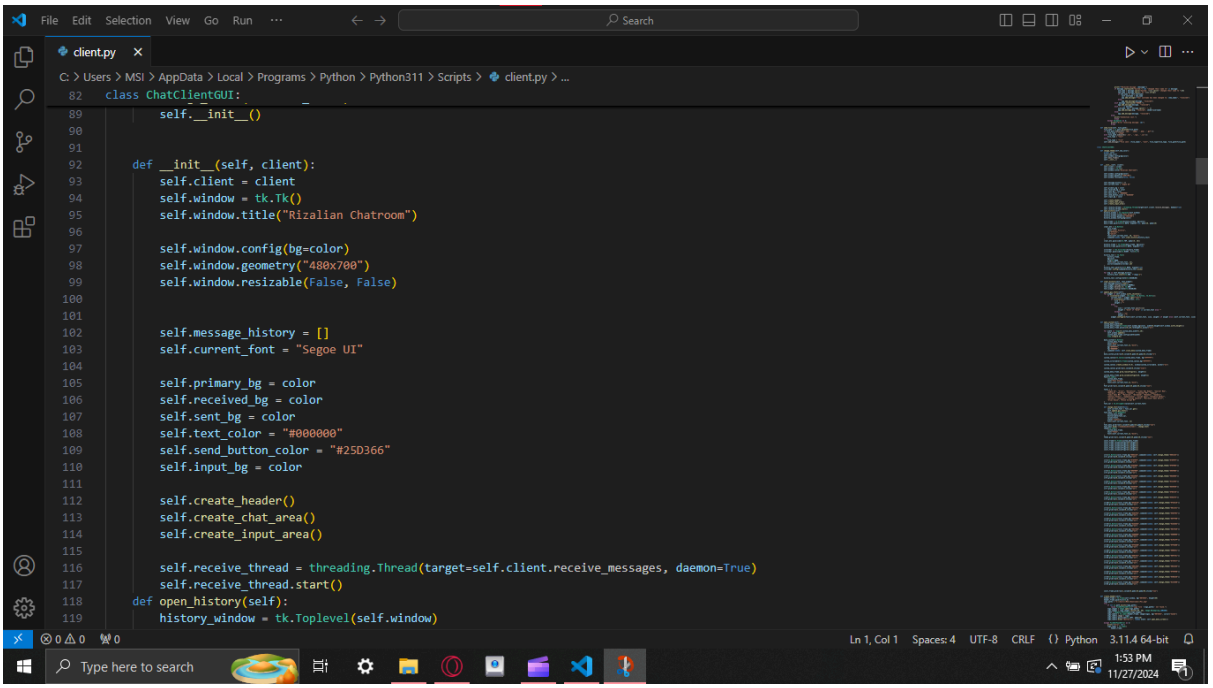
Create an intuitive and interactive chatroom interface with real-time communication, emoji support for engaging messaging, and scalability to handle multiple users simultaneously

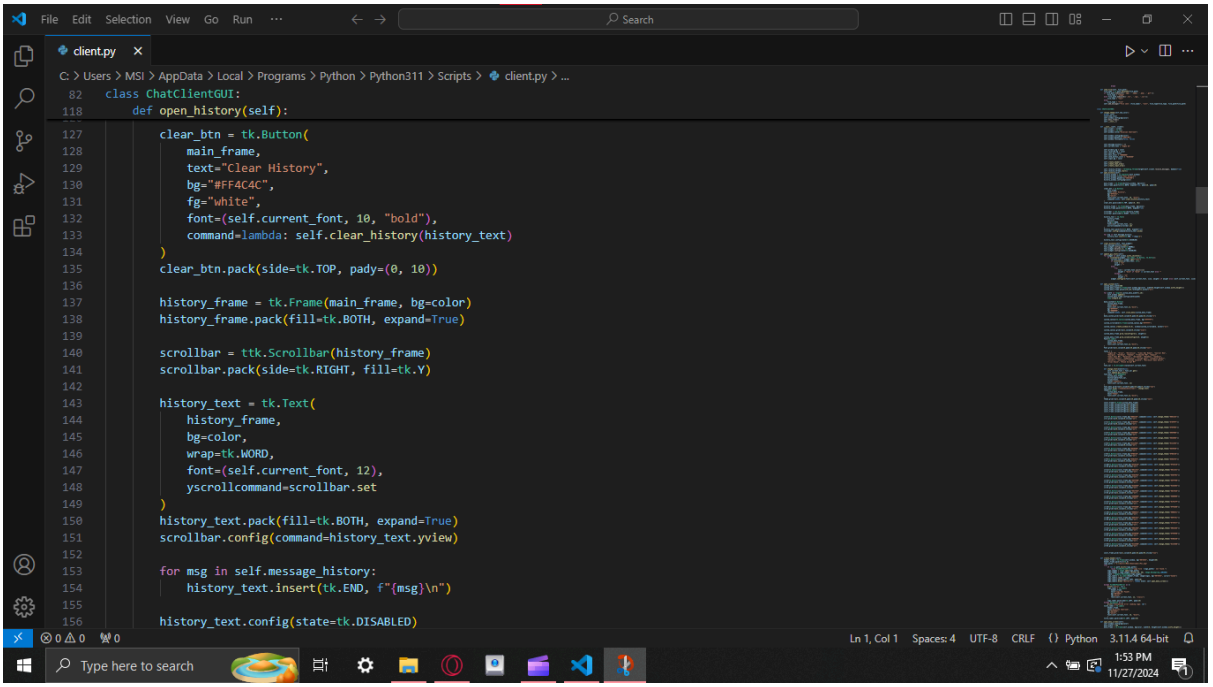
3. Screenshots of Source codes

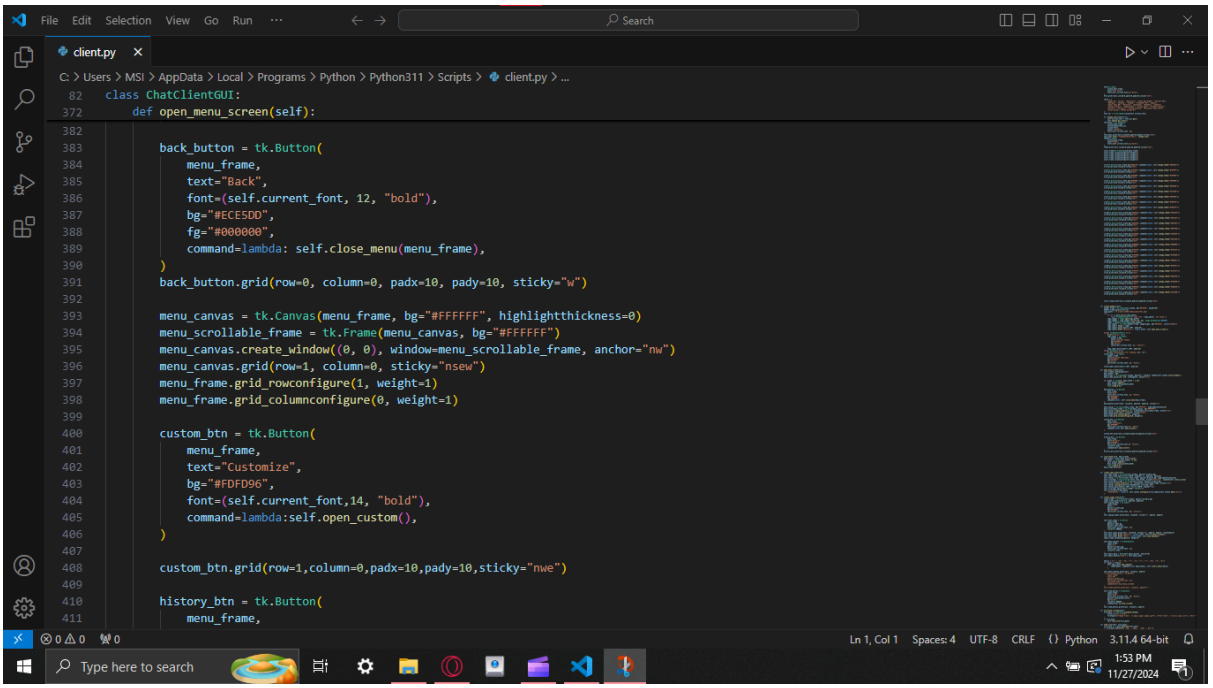
Client.py

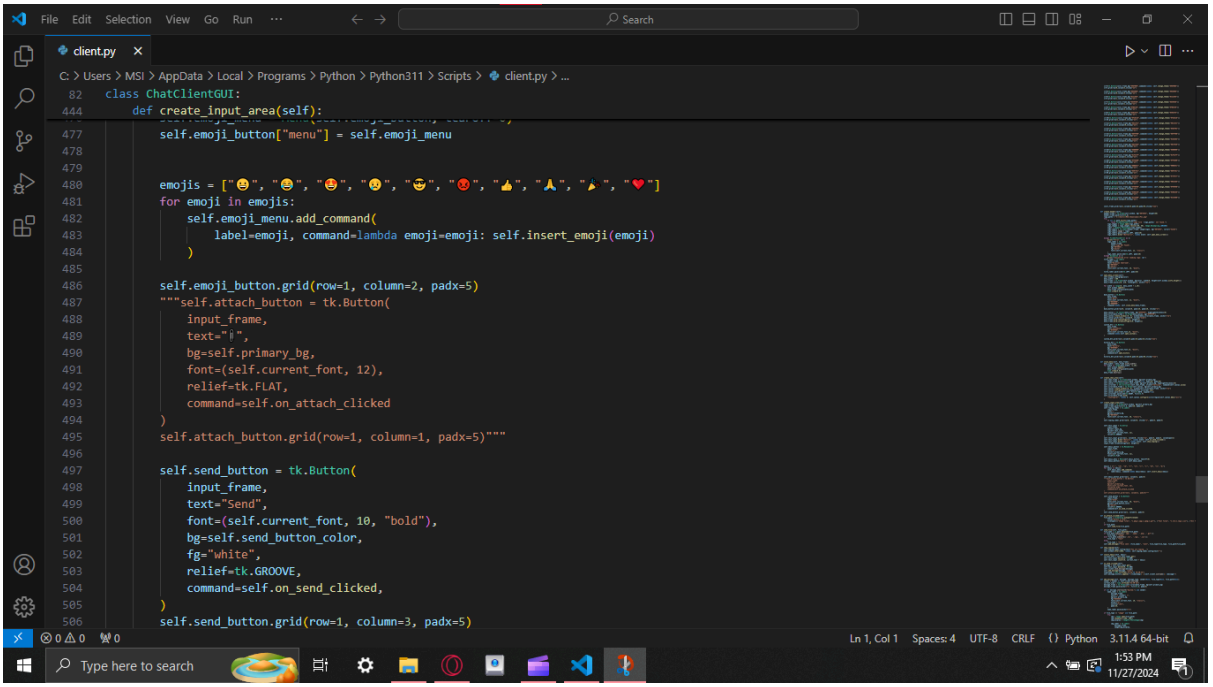


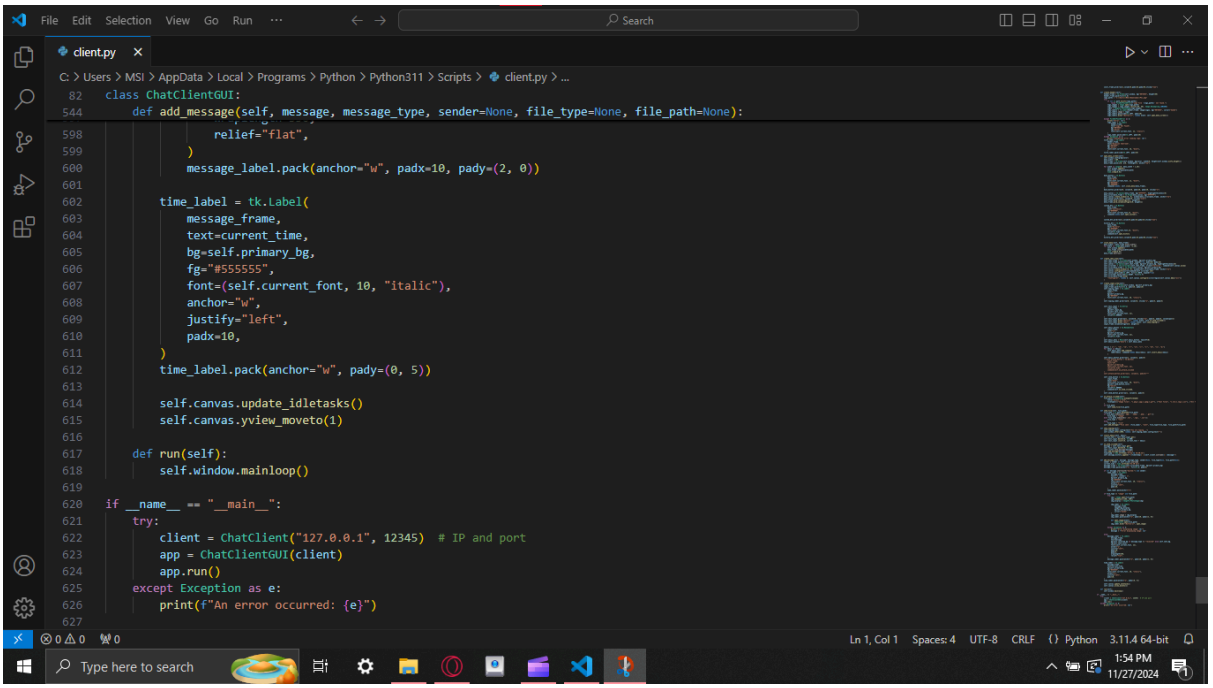

File Edit Selection View Go Run
Search
client.py
C: > Users > MSI > AppData > Local > Programs > Python > Python311 > Scripts > client.py > ...
class ChatClient:
def send_message(self, message):
try:
self.client_socket.send(message.encode())
except Exception as e:
print(f"Error sending message: {e}")
def receive_messages(self):
while True:
try:
message = self.client_socket.recv(1024).decode()
if message:
print(f"Received message: {message}")
if message.startswith("System:") and "changed their name to" in message:
old_name = message.split("System: ")[1].split(" changed their name to ")[0]
new_name = message.split(" to ")[1].strip()
if old_name == self.username:
self.username = new_name
app.add_message(f"Your username has been changed to: {new_name}", "received")
else:
app.add_message(message, "received")
elif message.startswith("ERROR:"):
app.add_message(message, "received")
elif ": " in message:
username, msg = message.split(": ", 1)
app.add_message(msg, "received", sender=username)
else:
app.add_message(message, "received")
else:
print("Connection lost.")
break
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Search
client.py
C: > Users > MSI > AppData > Local > Programs > Python > Python311 > Scripts > client.py > ...
class ChatClientGUI:
self.__init__()
def __init__(self, client):
self.client = client
self.window = tk.Tk()
self.window.title("Rizalian Chatroom")
self.window.config(bg=color)
self.window.geometry("480x700")
self.window.resizable(False, False)
self.message_history = []
self.current_font = "Segoe UI"
self.primary_bg = color
self.received_bg = color
self.sent_bg = color
self.text_color = "#000000"
self.send_button_color = "#25D366"
self.input_bg = color
self.create_header()
self.create_chat_area()
self.create_input_area()
self.receive_thread = threading.Thread(target=self.client.receive_messages, daemon=True)
self.receive_thread.start()
def open_history(self):
history_window = tk.Toplevel(self.window)
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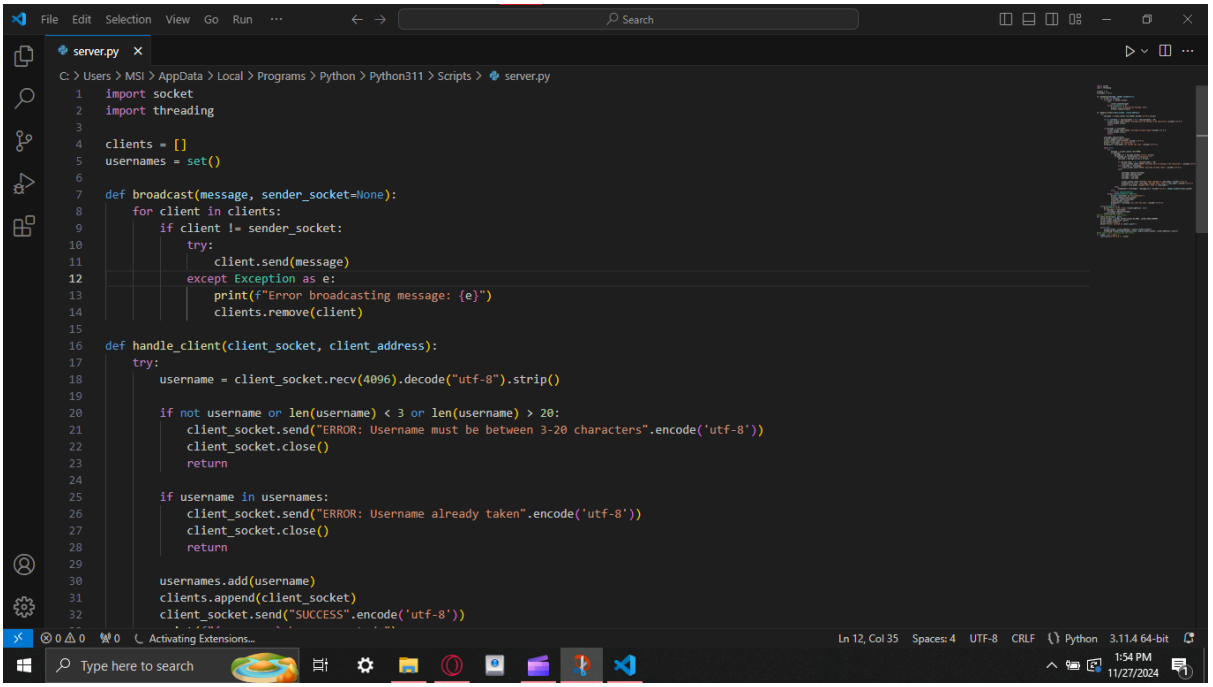

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client.py
C: > Users > MSI > AppData > Local > Programs > Python > Python311 > Scripts > client.py > ...
82 class ChatClientGUI:
118 def open_history(self):
127 clear_btn = tk.Button(
128 main_frame,
129 text="Clear History",
130 bg="#FF4C4C",
131 fg="white",
132 font=(self.current_font, 10, "bold"),
133 command=lambda: self.clear_history(history_text)
134)
135 clear_btn.pack(side=tk.TOP, pady=(0, 10))
136
137 history_frame = tk.Frame(main_frame, bg=color)
138 history_frame.pack(fill=tk.BOTH, expand=True)
139
140 scrollbar = ttk.Scrollbar(history_frame)
141 scrollbar.pack(side=tk.RIGHT, fill=tk.Y)
142
143 history_text = tk.Text(
144 history_frame,
145 bg=color,
146 wrap=tk.WORD,
147 font=(self.current_font, 12),
148 yscrollcommand=scrollbar.set
149)
150 history_text.pack(fill=tk.BOTH, expand=True)
151 scrollbar.config(command=history_text.yview)
152
153 for msg in self.message_history:
154 history_text.insert(tk.END, f"{msg}\n")
155
156 history_text.config(state=tk.DISABLED)
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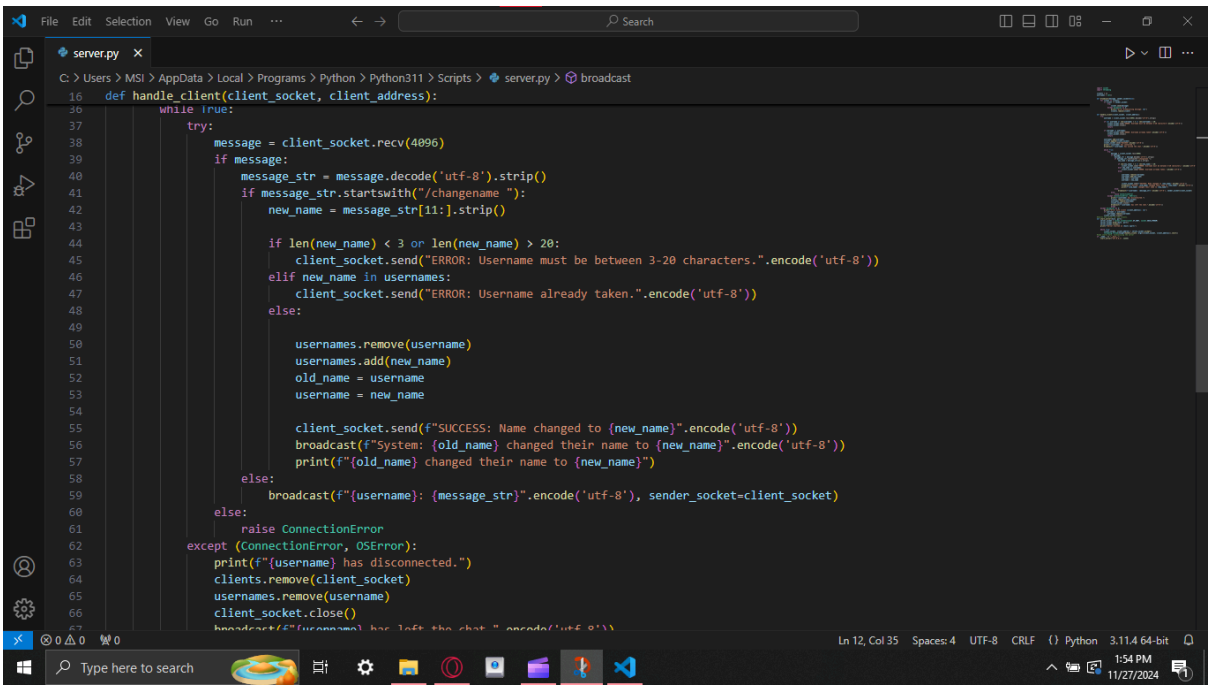

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client.py
C: > Users > MSI > AppData > Local > Programs > Python > Python311 > Scripts > client.py > ...
82 class ChatClientGUI:
372 def open_menu_screen(self):
382
383 back_button = tk.Button(
384 menu_frame,
385 text="Back",
386 font=(self.current_font, 12, "bold"),
387 bg="#ECE5DD",
388 fg="#000000",
389 command=lambda: self.close_menu(menu_frame),
390)
391 back_button.grid(row=0, column=0, padx=10, pady=10, sticky="w")
392
393 menu_canvas = tk.Canvas(menu_frame, bg="#FFFFFF", highlightthickness=0)
394 menu_scrollable_frame = tk.Frame(menu_canvas, bg="#FFFFFF")
395 menu_canvas.create_window((0, 0), window=menu_scrollable_frame, anchor="nw")
396 menu_canvas.grid(row=1, column=0, sticky="nsew")
397 menu_frame.grid_rowconfigure(1, weight=1)
398 menu_frame.grid_columnconfigure(0, weight=1)
399
400 custom_btn = tk.Button(
401 menu_frame,
402 text="Customize",
403 bg="#FDFD96",
404 font=(self.current_font,14, "bold"),
405 command=lambda:self.open_custom(),
406)
407
408 custom_btn.grid(row=1,column=0,padx=10,pady=10,sticky="nwe")
409
410 history_btn = tk.Button(
411 menu_frame,
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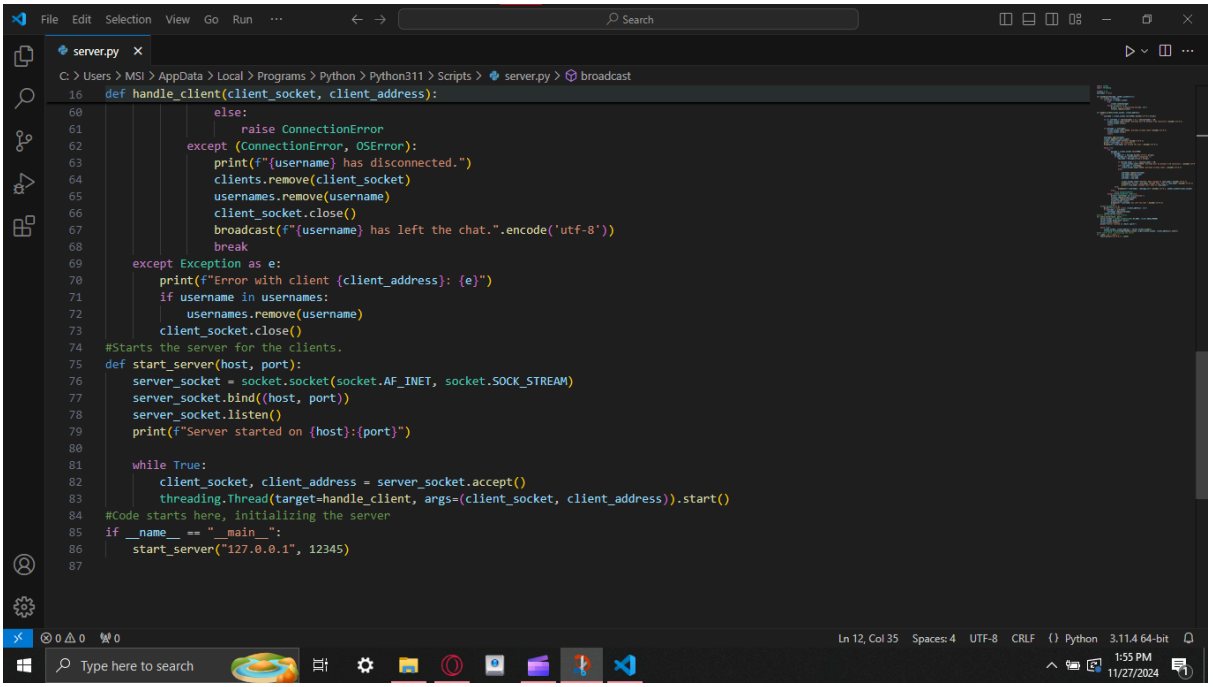

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client.py
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class ChatClientGUI:
def create_input_area(self):
self.emoji_button["menu"] = self.emoji_menu
emojis = ["😉", "😂", "🤩", "😢", "😎", "😡", "👍", "🙏", "🎉", "❤️"]
for emoji in emojis:
self.emoji_menu.add_command(
label=emoji, command=lambda emoji=emoji: self.insert_emoji(emoji)
)
self.emoji_button.grid(row=1, column=2, padx=5)
"""self.attach_button = tk.Button(
input_frame,
text="📎",
bg=self.primary_bg,
font=(self.current_font, 12),
relief=tk.FLAT,
command=self.on_attach_clicked
)
self.attach_button.grid(row=1, column=1, padx=5)"""
self.send_button = tk.Button(
input_frame,
text="Send",
font=(self.current_font, 10, "bold"),
bg=self.send_button_color,
fg="white",
relief=tk.GROOVE,
command=self.on_send_clicked,
)
self.send_button.grid(row=1, column=3, padx=5)
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client.py
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class ChatClientGUI:
def add_message(self, message, message_type, sender=None, file_type=None, file_path=None):
relief="flat",
)
message_label.pack(anchor="w", padx=10, pady=(2, 0))
time_label = tk.Label(
message_frame,
text=current_time,
bg=self.primary_bg,
fg="#555555",
font=(self.current_font, 10, "italic"),
anchor="w",
justify="left",
padx=10,
)
time_label.pack(anchor="w", pady=(0, 5))
self.canvas.update_idletasks()
self.canvas.yview_moveto(1)
def run(self):
self.window.mainloop()
if __name__ == "__main__":
try:
client = ChatClient("127.0.0.1", 12345) # IP and port
app = ChatClientGUI(client)
app.run()
except Exception as e:
print(f"An error occurred: {e}")
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Server.Py


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server.py
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import socket
import threading
clients = []
usernames = set()
def broadcast(message, sender_socket=None):
for client in clients:
if client != sender_socket:
try:
client.send(message)
except Exception as e:
print(f"Error broadcasting message: {e}")
clients.remove(client)
def handle_client(client_socket, client_address):
try:
username = client_socket.recv(4096).decode("utf-8").strip()
if not username or len(username) < 3 or len(username) > 20:
client_socket.send("ERROR: Username must be between 3-20 characters".encode('utf-8'))
client_socket.close()
return
if username in usernames:
client_socket.send("ERROR: Username already taken".encode('utf-8'))
client_socket.close()
return
usernames.add(username)
clients.append(client_socket)
client_socket.send("SUCCESS".encode('utf-8'))
Activating Extensions...
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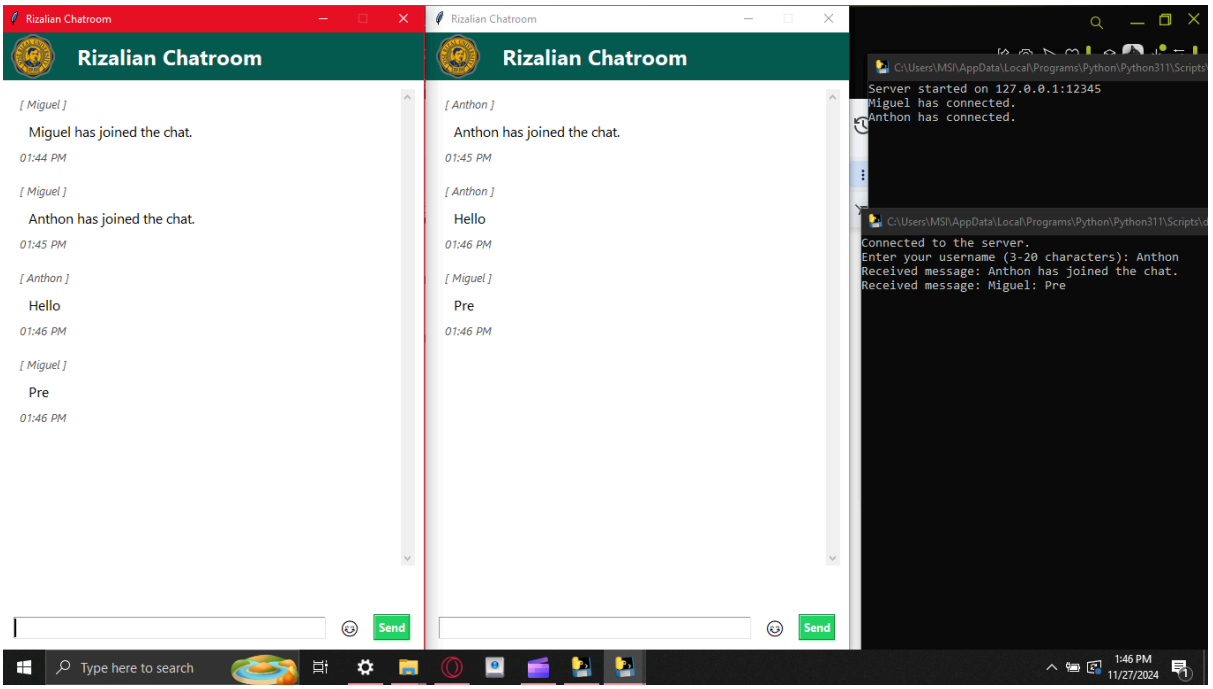

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Search
server.py
C: > Users > MSI > AppData > Local > Programs > Python > Python311 > Scripts > server.py > broadcast
def handle_client(client_socket, client_address):
while True:
try:
message = client_socket.recv(4096)
if message:
message_str = message.decode('utf-8').strip()
if message_str.startswith("/changename "):
new_name = message_str[11:].strip()
if len(new_name) < 3 or len(new_name) > 20:
client_socket.send("ERROR: Username must be between 3-20 characters.".encode('utf-8'))
elif new_name in usernames:
client_socket.send("ERROR: Username already taken.".encode('utf-8'))
else:
usernames.remove(username)
usernames.add(new_name)
old_name = username
username = new_name
client_socket.send(f"SUCCESS: Name changed to {new_name}".encode('utf-8'))
broadcast(f"System: {old_name} changed their name to {new_name}".encode('utf-8'))
print(f"{old_name} changed their name to {new_name}")
else:
broadcast(f"{username}: {message_str}".encode('utf-8'), sender_socket=client_socket)
else:
raise ConnectionError
except (ConnectionError, OSError):
print(f"{username} has disconnected.")
clients.remove(client_socket)
usernames.remove(username)
client_socket.close()
Ln 12, Col 35 Spaces: 4 UTF-8 CRLF Python 3.11.4 64-bit
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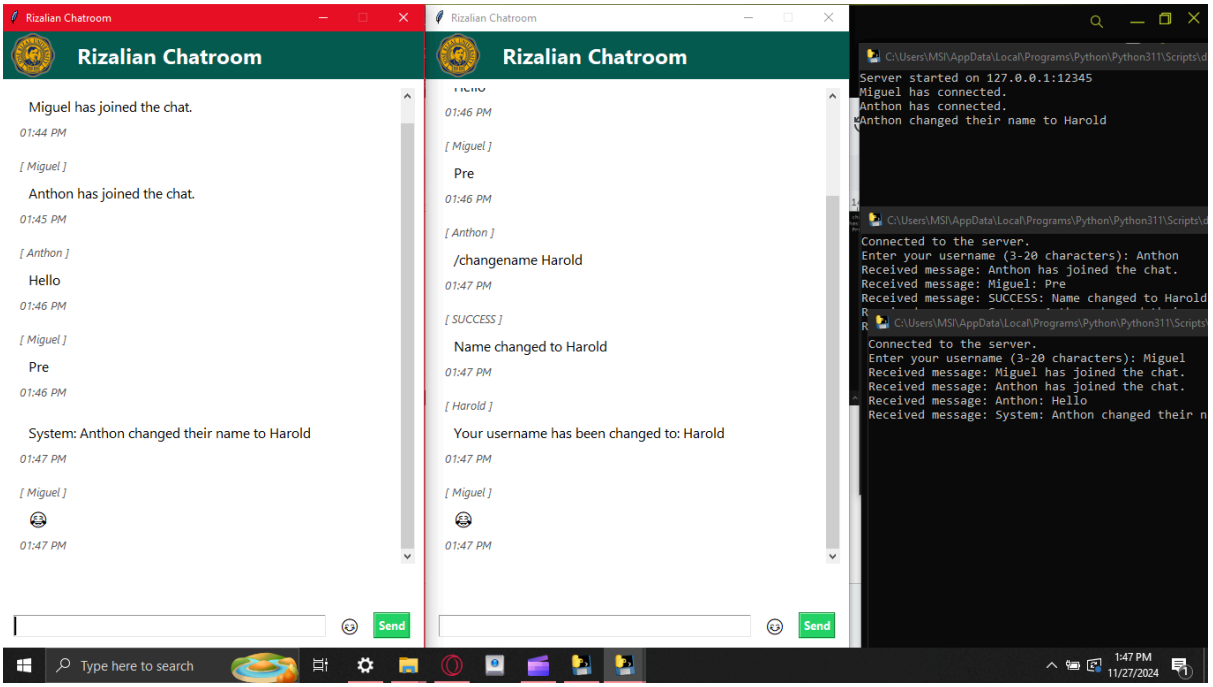


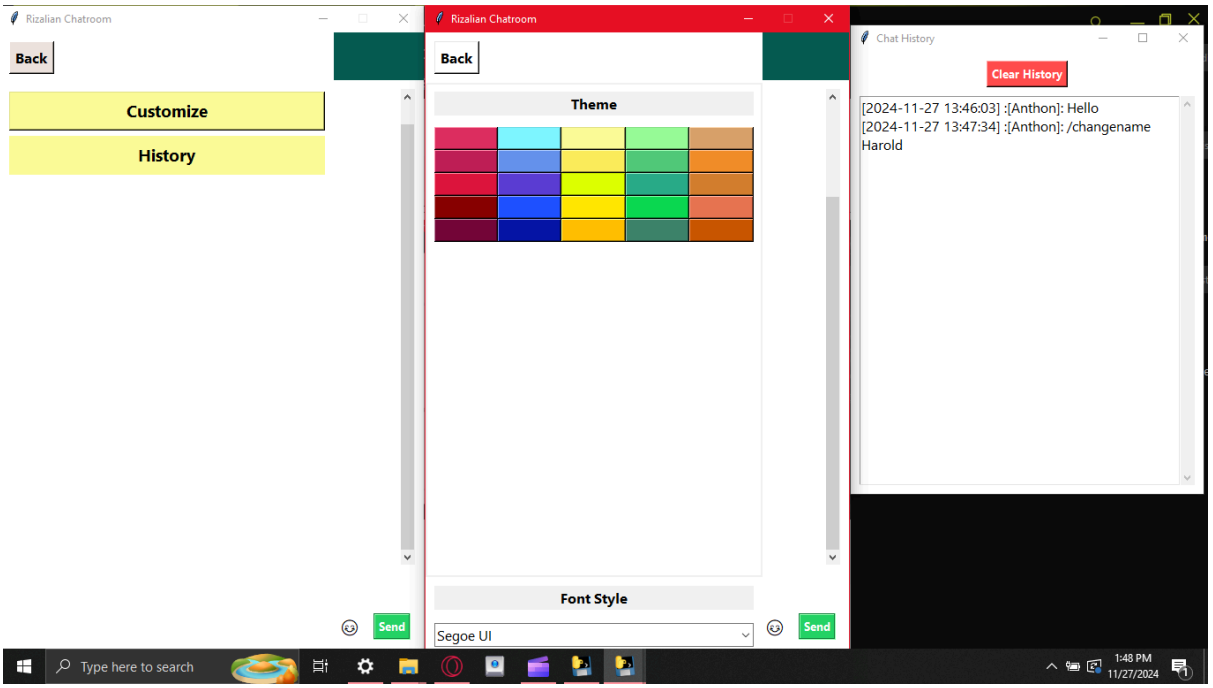
```
def handle_client(client_socket, client_address):
                    else:
                        raise ConnectionError
                except (ConnectionError, OSError):
                    print(f"{username} has disconnected.")
                    clients.remove(client_socket)
                    usernames.remove(username)
                    client_socket.close()
                    broadcast(f"{username} has left the chat.".encode('utf-8'))
                    break
        except Exception as e:
            print(f"Error with client {client_address}: {e}")
            if username in usernames:
                usernames.remove(username)
            client_socket.close()
    #Starts the server for the clients.
    def start_server(host, port):
        server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        server_socket.bind((host, port))
        server_socket.listen()
        print(f"Server started on {host}:{port}")

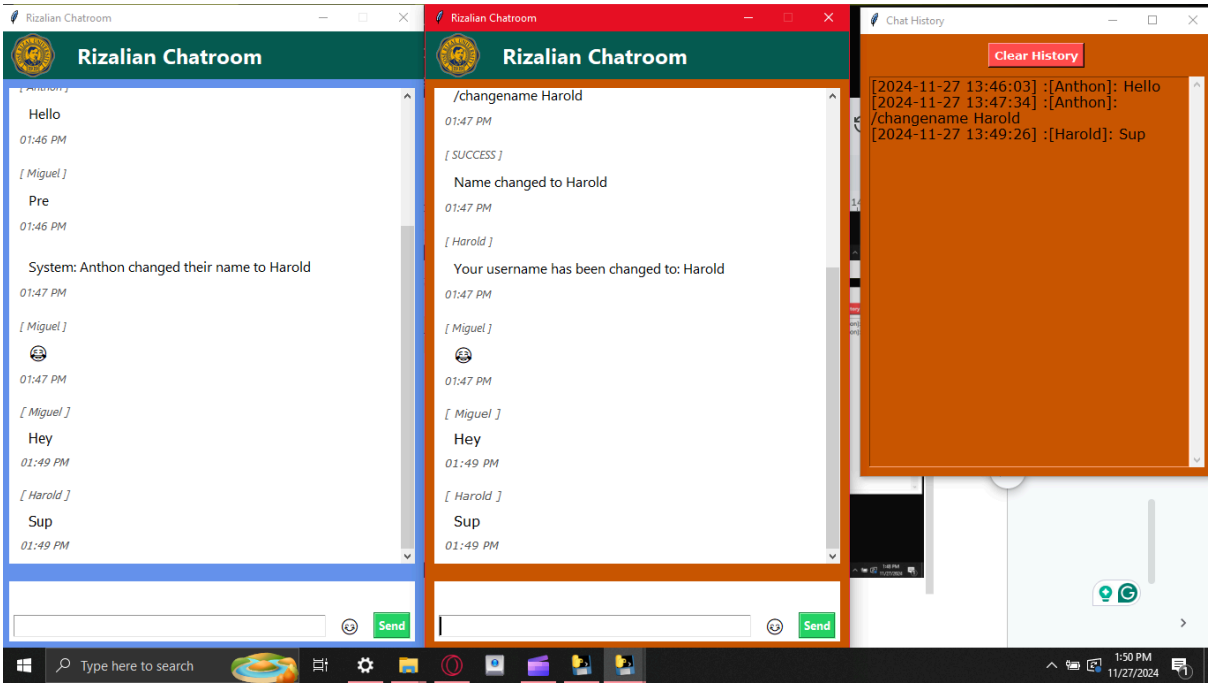
        while True:
            client_socket, client_address = server_socket.accept()
            threading.Thread(target=handle_client, args=(client_socket, client_address)).start()
    #Code starts here, initializing the server
    if __name__ == "__main__":
        start_server("127.0.0.1", 12345)
```

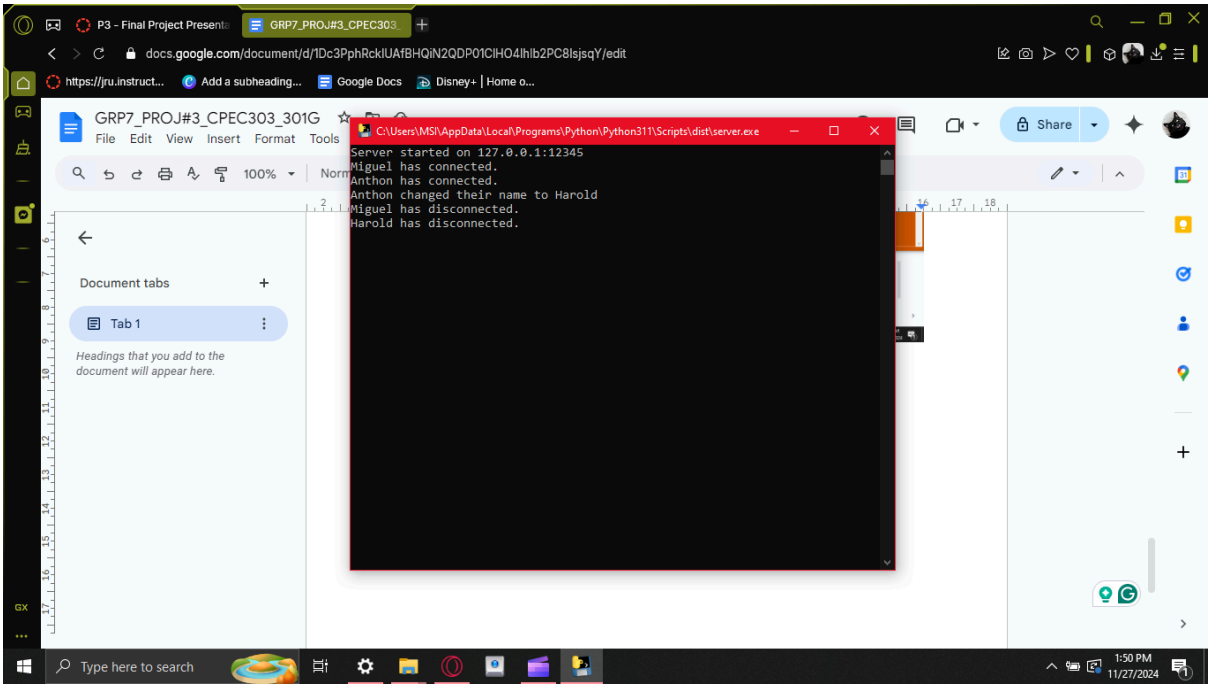
4. Screenshots of Project Testing/Running




Rizalian Chatroom
Rizalian Chatroom
Miguel has joined the chat.
01:44 PM
[Miguel]
Anthon has joined the chat.
01:45 PM
[Anthon]
Hello
01:46 PM
[Miguel]
Pre
01:46 PM
System: Anthon changed their name to Harold
01:47 PM
[Miguel]
01:47 PM
Send
Rizalian Chatroom
Rizalian Chatroom
01:46 PM
[Miguel]
Pre
01:46 PM
[Anthon]
/changename Harold
01:47 PM
[SUCCESS]
Name changed to Harold
01:47 PM
[Harold]
Your username has been changed to: Harold
01:47 PM
[Miguel]
01:47 PM
Send
Server started on 127.0.0.1:12345
Miguel has connected.
Anthon has connected.
Anthon changed their name to Harold
Connected to the server.
Enter your username (3-20 characters): Anthon
Received message: Anthon has joined the chat.
Received message: Miguel: Pre
Received message: SUCCESS: Name changed to Harold
Connected to the server.
Enter your username (3-20 characters): Miguel
Received message: Miguel has joined the chat.
Received message: Anthon has joined the chat.
Received message: Anthon: Hello
Received message: System: Anthon changed their n
Type here to search
1:47 PM
11/27/2024


Rizalian Chatroom
Back
Customize
History
Send
Rizalian Chatroom
Back
Theme
Font Style
Segoe UI
Send
Chat History
Clear History
[2024-11-27 13:46:03] :[Anthon]: Hello
[2024-11-27 13:47:34] :[Anthon]: /changename Harold
Type here to search
1:48 PM
11/27/2024


Rizalian Chatroom
Hello
01:46 PM
[Miguel]
Pre
01:46 PM
System: Anthon changed their name to Harold
01:47 PM
[Miguel]
01:47 PM
[Miguel]
Hey
01:49 PM
[Harold]
Sup
01:49 PM
Send
Rizalian Chatroom
/changename Harold
01:47 PM
[SUCCESS]
Name changed to Harold
01:47 PM
[Harold]
Your username has been changed to: Harold
01:47 PM
[Miguel]
01:47 PM
[Miguel]
Hey
01:49 PM
[Harold]
Sup
01:49 PM
Send
Chat History
Clear History
[2024-11-27 13:46:03] :[Anthon]: Hello
[2024-11-27 13:47:34] :[Anthon]:
/changename Harold
[2024-11-27 13:49:26] :[Harold]: Sup
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1:50 PM
11/27/2024


P3 - Final Project Present
GRP7_PROJ#3_CPEC303
docs.google.com/document/d/1Dc3PphRcklUAfBHQiN2QDP01ClHO4lhlb2PC8lsjsqY/edit
https://jru.instruct...
Add a subheading...
Google Docs
Disney+ | Home o...
GRP7_PROJ#3_CPEC303_301G
File Edit View Insert Format Tools
Share
C:\Users\MSI\AppData\Local\Programs\Python\Python311\Scripts\dist\server.exe
Server started on 127.0.0.1:12345
Miguel has connected.
Anthon has connected.
Anthon changed their name to Harold
Miguel has disconnected.
Harold has disconnected.
Document tabs
Tab 1
Headings that you add to the document will appear here.
Type here to search
1:50 PM
11/27/2024